AI-Based Anomaly Detection and Intelligent Monitoring of Sewage Treatment Plants: A Literature Survey and Proposed Research Direction

Amisha Kirana Kumar
Department of AIML,
BMS Institute of Technology and Management,
Bengaluru.India
25ug1byai148@bmsit.in

Deepak Bandi
Department of AIML,
BMS Institute of Technology and Management,
Bengaluru.India
25ug1byai208@bmsit.in

Nikhilkumar
Department of AIML,
BMS Institute of Technology and Management,
Bengaluru.India
25ug1byai243@bmsit.in

Vibha V
Department of AIML,
BMS Institute of Technology and Management,
Bengaluru.India
vibha00813@gmail.com

I. INTRODUCTION

Sewage Treatment Plants (STPs) play an important role in treating domestic and industrial wastewater before its discharge or reuse. Effective monitoring of STP operation depends on reliable measurements from sensors that monitor important water-quality parameters. However, sensor readings may sometimes contain abnormal values because of sensor faults, noise, calibration problems, environmental changes, or other unexpected conditions.

Artificial Intelligence (AI) and Machine Learning (ML) techniques provide opportunities to analyze large amounts of sensor data automatically and identify unusual patterns. Recent research has demonstrated the use of AI-based soft sensors for estimating wastewater-quality parameters such as Total Suspended Solids (TSS), Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Nitrogen (TN), and Total Phosphorus (TP) from turbidity measurements. However, prediction of water-quality parameters and detection of abnormal sensor readings are different objectives.

Therefore, this project proposes an AI-Based Detection of Abnormal Sensor Readings in Karnataka Sewage Treatment Plants. The proposed system focuses on identifying sensor observations that deviate from normal operating patterns. By detecting such abnormal readings at an early stage, the system can support STP operators in monitoring sensor reliability, identifying readings that require verification, and improving the effectiveness of wastewater-treatment monitoring.operating patterns. By detecting such abnormal readings at an early stage, the system can support STP operators in monitoring sensor reliability, identifying readings that require verification, and improving the effectiveness of wastewater-treatment monitoring.

II. SYSTEMATIC LITERATURE REVIEW METHODOLOGY

The systematic literature review involves collecting and studying research papers related to AI-based anomaly detection and intelligent monitoring of wastewater and sewage treatment plants. The selected papers are analysed based on the AI techniques used, sensor data, types of anomalies detected, explainability, performance, and real-world applicability. The findings are then compared to identify the limitations and research gaps in existing systems, which helps to define the proposed research direction.

TABLE I
LITERATURE REVIEW OF AI-BASED ANOMALY DETECTION AND INTELLIGENT MONITORING STUDIES

No.	Paper Details	Objective	Parameters / Data	AI/ML Method	Limitation / Research Gap
1	"Wastewater treatment monitoring: Fault detection in sensors using transductive learning and improved reinforcement learning" (2026)	Detect faults in WWTP sensors.	Time-series sensor measurements from oxidation and nitrification processes.	Spatial-attention Transductive LSTM + Reinforcement Learning	Advanced model; lightweight deployment and interpretability require further study.
2	"Dual-stage soft sensor-based fault reconstruction and effluent prediction toward a sustainable wastewater treatment plant using attention fusion deep learning model" (2026)	Reconstruct faulty sensor readings and predict effluent quality.	Dissolved oxygen and effluent COD.	Attention-Fusion Autoencoder + Attention CNN	Multi-STP evaluation is required for broader applicability.
3	"Data-driven fault detection and diagnosis methods in wastewater treatment systems: A comprehensive review" (2026)	Review data-driven fault detection and diagnosis methods in wastewater treatment.	Process and sensor data.	Statistical, ML and hybrid FDD methods.	Review-based study; does not develop a new multi-STP detection system.
4	"Effluent quality soft sensor for wastewater treatment plant with ensemble sparse learning-based online next generation reservoir computing" (2025)	Develop an online soft sensor for real-time wastewater-quality monitoring.	Effluent quality variables including COD/BOD-related parameters.	Next Generation Reservoir Computing + Ensemble Sparse Learning	Mainly focuses on prediction rather than direct abnormal sensor detection.
5	"Learning a neural network-based soft sensor with double-errors parallel optimization towards effluent variable prediction in wastewater treatment plants"(2024)	Improve prediction of important effluent variables.	Effluent-quality variables.	Neural Network + Double-Errors Parallel Optimization	Prediction-focused; anomaly detection and cross-plant deployment require further study.
6	"Machine learning methods for anomaly classification in wastewater treatment plants" (2023)	Detect and classify abnormal WWTP operating conditions.	Oxygen, ammonia and aeration-related parameters.	SVM, MLP, Random Forest, LightGBM and XGBoost	Performance may depend on specific plant/process conditions; generalization remains a challenge.
7	"A novel long short-term memory artificial neural network (LSTM)-based soft-sensor to monitor and forecast wastewater treatment performance" (2023)	Predict and monitor WWTP performance.	DO, ORP, suspended solids, COD, NH_4^+ and TN.	LSTM + Multiple Linear Regression	Focuses mainly on prediction/forecasting rather than direct sensor-anomaly detection.
8	"Intelligent sensor validation for sustainable influent quality monitoring in wastewater treatment plants using stacked denoising autoencoders" (2021)	Detect faulty sensor measurements and perform sensor validation/reconciliation.	Influent-quality sensor measurements.	Stacked Denoising Autoencoder	Further validation across different plants and operating conditions is needed.
9	"Advances in soft sensors for wastewater treatment plants: A systematic review". (2021)	Review the development and application of soft sensors for WWTP monitoring.	Wastewater process and sensor data.	Statistical methods, ML, neural networks and tree-based methods.	Highlights the need for reliable online monitoring and broader practical deployment of soft sensors.
10	"Incipient fault detection of sensors used in wastewater treatment plants based on deep dropout neural network" (2020)	Detect early-stage faults in WWTP sensors.	DO, pH and MLSS sensor data.	Deep Dropout Neural Network	Validation across a wider range of real-world WWTPs is required.

III. OBJECTIVES

1. To review AI and data-driven techniques used for anomaly detection and intelligent monitoring of WWTPs.
2. To critically analyse existing approaches with respect to data characteristics, anomaly types, explainability, generalization and real-world applicability.
3. To identify the research gap and propose a plant-context-aware, explainable multi-parameter anomaly detection direction for heterogeneous STPs.

IV. AI-BASED WWTP MONITORING AND ANOMALY DETECTION

AI-based techniques are increasingly used in wastewater treatment plants (WWTPs) to monitor process conditions, identify abnormal sensor readings, and improve treatment performance. The main approaches include statistical methods, machine learning, deep learning, and explainable AI.

Statistical Methods: Statistical techniques use the normal behaviour and statistical patterns of sensor data to identify unusual observations. Methods such as Principal Component Analysis (PCA), control charts, correlation analysis, and threshold-based detection can be used to identify deviations in parameters such as pH, dissolved oxygen (DO), turbidity, flow, and temperature. These methods are relatively simple and computationally efficient, but they may have difficulty detecting complex nonlinear relationships between multiple sensor parameters.

Machine Learning: Machine learning methods learn patterns from historical sensor data and use them to identify abnormal or faulty conditions. Algorithms such as Isolation Forest, Support Vector Machine (SVM), Random Forest, K-Means, and XGBoost can be applied for anomaly and fault detection. These methods can handle multiple sensor parameters and are useful when abnormal conditions are difficult to identify using fixed thresholds.

Deep Learning: Deep learning techniques are useful for large-scale and time-series wastewater data. Models such as Autoencoders, Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) can learn complex temporal and nonlinear patterns in sensor readings. Autoencoders can identify anomalies by measuring reconstruction errors, while LSTM models can learn changes in sensor behaviour over time. However, deep-learning models generally require more data and computational resources.

Explainable AI: Explainable AI (XAI) helps users understand why an AI model identifies a sensor reading as abnormal. Techniques such as SHAP and feature importance can indicate which parameters contributed most to an anomaly. For example, an abnormal DO reading may be associated with changes in flow, temperature, or turbidity. XAI can therefore improve operator trust and support faster diagnosis and corrective action in WWTPs.

TABLE II
COMPARISON OF AI-BASED APPROACHES

Approach	Examples	Main Use	Advantage	Limitation
Statistical	PCA, control charts, thresholds	Detect deviations	Simple and fast	Limited for nonlinear patterns
Machine Learning	Isolation Forest	Anomaly/fault detection	Handles multiple variables	Needs suitable training data
Deep Learning	Autoencoder, LSTM, CNN	Complex/time-series anomalies	Learns complex patterns	Data and computation intensive
Explainable AI	SHAP, feature importance	Explain detected anomalies	Improves interpretability	Adds extra processing/model complexity

V. CRITICAL COMPARATIVE ANALYSIS OF EXISTING STUDIES

Existing studies were compared based on their datasets, data sources, monitored parameters, preprocessing methods, AI/ML techniques, anomaly or fault types, detection approaches, explainability, validation methods, and performance measures. The studies were also examined for real-time monitoring capability, robustness, computational requirements, and applicability across different wastewater treatment plants. Most existing approaches focus on specific sensor faults, process anomalies, or wastewater-quality prediction, while some methods provide limited explanation of why an observation is classified as abnormal. In addition, many studies are evaluated using data from a single plant or a limited set of operating conditions. These limitations indicate the need for a practical and interpretable anomaly-detection approach that can handle multiple sensor parameters and support reliable monitoring across sewage treatment plant

TABLE III
COMPARISON POINTS FOR EXISTING STUDIES

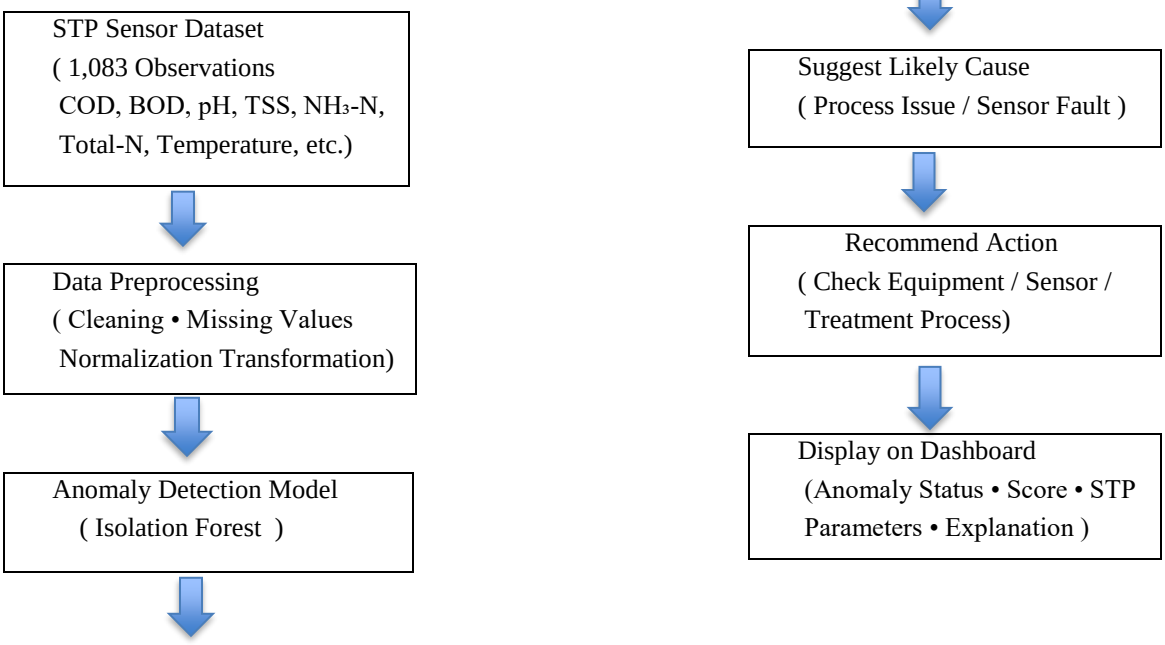
Comparison Point	What to Analyze
1. Dataset	Source of data, real plant/simulation, dataset size, time period
2. Parameters	pH, DO, COD, BOD, TSS, turbidity, flow, temperature, ORP, ammonia, etc.
3. Preprocessing	Missing-value handling, normalization, filtering, outlier removal, resampling
4. Method	Statistical, ML, DL, hybrid, or ensemble techniques
5. Anomaly/Fault Type	Sensor fault, process fault, abnormal measurement, drift, bias, noise, missing data
6. Explainability	SHAP, feature importance, reconstruction error, rule-based explanation
7. Validation	Train/test split, cross-validation, real plant validation, simulation
8. Alert/Decision Support	Whether the study provides warnings, severity levels, or recommended actions
9. Limitations	Dataset size, single-plant evaluation, lack of explainability, high complexity, etc.

VI. RESEARCH GAP

Existing WWTP research has extensively investigated sensor/process fault detection, anomaly detection, fault reconstruction, and predictive modelling. However, limited attention has been given to integrating multi-parameter anomaly detection with plant-specific characteristics such as treatment technology, capacity, environmental conditions and historical operating behaviour, while also providing interpretable parameter-level explanations across multiple heterogeneous STPs.

VII. PROPOSED RESEARCH DIRECTION

The proposed research focuses on developing an AI-based, explainable anomaly detection system for Sewage Treatment Plants (STPs). The system will analyse multiple sensor readings such as pH, DO, temperature, TSS, turbidity, and flow to detect abnormal patterns. It will also identify the affected sensor and location, explain why the reading is abnormal, suggest a likely cause, and provide a recommended action to help STP operators respond quickly and effectively.



VIII. CONCLUSION

The proposed research focuses on using AI-based anomaly detection to improve monitoring of Sewage Treatment Plants (STPs). The system analyses multiple sensor parameters such as pH, DO, temperature, TSS, turbidity, and flow to detect abnormal readings. It identifies the affected sensor, location, and time and provides an understandable explanation of the detected anomaly. A rule-based approach can be used to suggest the possible cause and recommended action. This approach can help STP operators detect problems early, reduce manual monitoring, and improve maintenance. Future work can focus on testing the system using larger real-world datasets from multiple STPs.

References

[1] J. Yang, K. Tian, H. Zhao, Z. Feng, S. Bourouis, S. Dhahbi, A. A. Khan, M. Berrima, and L. Y. Por, "Wastewater treatment monitoring: Fault detection in sensors using transductive learning and improved reinforcement learning," *Expert Systems with Applications*, vol. 264, Art. no. 125805, 2025, doi: 10.1016/j.eswa.2024.125805.

[2] A. H. Ba-Alawi and J. Kim, "Dual-stage soft sensor-based fault reconstruction and effluent prediction toward a sustainable wastewater treatment plant using attention fusion deep learning model," *Journal of Environmental Chemical Engineering*, vol. 13, no. 3, Art. no. 116221, 2025, doi: 10.1016/j.jece.2025.116221.

[3] X. Wang, W. Wei, C. Yoo, and H. Liu, "Data-driven fault detection and diagnosis methods in wastewater treatment systems: A comprehensive review," *Environmental Research*, vol. 268, Art. no. 120822, 2025, doi: 10.1016/j.envres.2025.120822.

[4] G. Fang, D. Huang, and Y. Liu, "Effluent quality soft sensor for wastewater treatment plant with ensemble sparse learning-based online next generation reservoir computing," *Water Research X*, vol. 25, Art. no. 100276, 2024, doi: 10.1016/j.wroa.2024.100276.

[5] D. Li, C. Yang, Y. Li, Y. Chen, D. Huang, and Y. Liu, "Learning a neural network-based soft sensor with double-errors parallel optimization towards effluent variable prediction in wastewater treatment plants," *Journal of Environmental Management*, vol. 366, Art. no. 121907, 2024, doi: 10.1016/j.jenvman.2024.121907.

[6] "Machine learning methods for anomaly classification in wastewater treatment plants," *Journal of Environmental Management*, vol. 344, Art. no. 118594, 2023, doi: 10.1016/j.jenvman.2023.118594.

[7] B. Xu, C. K. Pooi, K. M. Tan, S. Huang, X. Shi, and H. Y. Ng, "A novel long short-term memory artificial neural network (LSTM)-based soft-sensor to monitor and forecast wastewater treatment performance," *Journal of Water Process Engineering*, vol. 54, Art. no. 104041, 2023, doi: 10.1016/j.jwpe.2023.104041.

[8] A. H. Ba-Alawi, P. Vilela, J. Loy-Benitez, S. K. Heo, and C. K. Yoo, "Intelligent sensor validation for sustainable influent quality monitoring in wastewater treatment plants using stacked denoising autoencoders," *Journal of Water Process Engineering*, vol. 43, Art. no. 102206, 2021, doi: 10.1016/j.jwpe.2021.102206.

[9] P. M. L. Ching, R. H. Y. So, and T. Morck, "Advances in soft sensors for wastewater treatment plants: A systematic review," *Journal of Water Process Engineering*, vol. 44, Art. no. 102367, 2021, doi: 10.1016/j.jwpe.2021.102367.

[10] B. Mali and S. H. Laskar, "Incipient fault detection of sensors used in wastewater treatment plants based on deep dropout neural network," *SN Applied Sciences*, vol. 2, Art. no. 2121, 2020, doi: 10.1007/s42452-020-03910-9.